

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Course Name:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|   | Microprocessor and Microcontroller Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2 | <b>Course Code:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|   | IBB20204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3 | <b>Year and Semester Offered:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|   | Year 2 Semester 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4 | <b>Credit Hours Value:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|   | 4 Credit Hours [4 SLT Credit]                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5 | <b>Prerequisite(s), if applicable:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|   | IBB30104 Digital Principles                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6 | <b>Course Synopsis:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|   | This course introduces the theory behind computers and other digital instruments. The microprocessor chapters use the 8086 microprocessors to explain the fundamentals of microprocessor architecture, programming and hardware. In this module, students will learn about the linkage between the digital electronic theory and the hardware/software requirement to interface to real applications via microprocessor ICs.                                                                 |
| 7 | <b>Course Learning Outcome:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|   | <p>Upon completion of the course, students should be able to:</p> <ol style="list-style-type: none"> <li>1. Describe the microprocessor architecture, its instruction set and addressing modes</li> <li>2. Explain instruction execution relates to memory organization and memory interfacing.</li> <li>3. Examine and develop a simple assembly language program</li> <li>4. Demonstrate microprocessor's operations by using simulator or tools for microprocessor application</li> </ol> |
| 8 | <b>Content Outline:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|   | <b>Topic</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|   | 1. Introduction to Microprocessor and Microcontroller                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|   | 2. Architecture of the 8086 Microprocessor                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|   | 3. Memory Organization and Interfacing                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|   | 4. Instruction Set and Addressing Modes of 8086 Microprocessor                                                                                                                                                                                                                                                                                                                                                                                                                               |
|   | 5. Introduction to Assembly Language                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|   | 6. Programming of 8086 Microprocessor                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|   | 7. Application of Microprocessor and Microcontroller                                                                                                                                                                                                                                                                                                                                                                                                                                         |